Tables of Large Sets LS$[N](t,k,v)$ of Prime Size for $2 \le t \le 6$ and $v \le 100$

Supplement to "New Large Sets of t-Designs from Affine Groups"

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Scope and conventions

The tables cover $N \in \{2,3,5,7,11,13,17,19,23,29,31,37,41,43\}$, $2 \le t \le 6$, $t < k \le v/2$ and $v \le 100$. A parameter set $(N;t,k,v)$ is *admissible* when N divides $\binom{v-i}{k-i}$ for $0 \le i \le t$. For each N and t the table has one row for each v with an admissible k. The first column lists the values of k for which a large set exists, with the values for which existence follows from the results of the accompanying paper in bold, and the second column lists the admissible values that are open. The parameter sets for which a large set is known not to exist are listed above each table. Parameter sets with $v - k < t$ are omitted.

Every entry marked as existing is derived by a program from a list of base results by the recursive theorems listed below. The file `existence_table.csv` gives a justification for each of the 50441 admissible parameter sets: its source, or the rule and the parameter sets from which it was derived.

Recursive theorems used

1. All admissible large sets of 1-designs exist (Baranyai, Hartman).
2. An LS$[N](t,k,v)$ gives an LS$[N](t-i,k-j,v-l)$ for $0 \le j \le l \le i \le t$, and an LS$[N](t,v-k,v)$.
3. An LS$[N](t,k,v)$ and an LS$[N](t,k+1,v)$ give an LS$[N](t,k+1,v+1)$ (Ajoodani-Namini and Khosrovshahi).
4. If LS$[N](t,i,v)$ exist for $t < i \le k$ and an LS$[N](t,k,u)$ exists, then LS$[N](t,k,u+l(v-t))$ exist for $l \ge 1$ (Ajoodani-Namini and Khosrovshahi).
5. For a prime p, an LS$[p](t,k,v-1)$ gives LS$[p](t+1,pk+i,pv+j)$ for $0 \le j < i \le p-1$ (Ajoodani-Namini) and LS$[p](t,pk+i,pv+j)$ for $-p \le j < i \le p-1$ (Ajoodani-Namini, Tayfeh-Rezaie).
6. An LS$[N](t,k,2k+1)$ with t even, or with $N = 2$, gives an LS$[N](t+1,k+1,2k+2)$ (Alltop).
7. For a prime p: if LS$[p](t,i,p^f+t)$ exist for $t < i \le \min(k,(p^f+t)/2)$ and $t \le v \bmod p^f < k \bmod p^f$, then an LS$[p](t,k,v)$ exists (Khosrovshahi and Tayfeh-Rezaie).

Non-existence is propagated through item 2.

Base results

General theorems: $t = 2$, $k = 3$, all N, $v \ne 7$ (Lu, Teirlinck). $N = 2$: $t = 2$ and all k (Ajoodani-Namini), $3 \le t \le 5$ and $k \le 15$, $t = 6$ and $k \in \{7,8,9\}$. $N = 3$: $t \in \{2,3\}$ and $k \le 80$, $t = 4$ and $k \le 10$. $N = 5$: $t = 2$, $k \le 8$, and $t = 3$, $k \le 6$. $N = 7$: $t \in \{2,3\}$, $k \le 6$. $N = 11$: $t = 2$, $k \le 10$. $N \in \{13,17,19,23\}$: $t = 2$, $k \in \{3,4\}$. $N = 29$: $t = 2$, $k \le 28$. Minimum-index large sets of 3-$(v,4,\lambda)$ designs for $v \equiv 0 \pmod 3$ (Teirlinck).

Directly constructed large sets, with the keys used in the CSV file: KMS91 (Kramer, Magliveras, Stinson, Australas. J. Combin. 3 (1991) 191–205), Chouinard83 (Discrete Math. 45 (1983) 297–300), KMO93 (Kramer, Magliveras, O'Brien, Australas. J. Combin. 7 (1993) 189–193), CCFK90 (Chee, Colbourn, Furino, Kreher, Australas. J. Combin. 2 (1990) 111–119), CM98 (Chee, Magliveras, J. Combin. Des. 6 (1998) 293–308), Denniston83 (Ann. Discrete Math. 18 (1983) 291–294), Eslami03 (J. Combin. Des. 11 (2003) 312–316), CuMa99 (Cusack, Magliveras, Des. Codes Cryptogr. 18 (1999) 81–87), LMW01 (Laue, Magliveras, Wassermann, J. Combin. Des. 9 (2001) 40–59, including the large sets obtained from its Theorem 5 for affine and projective groups with orbits of two lengths), LOTW07 (Laue, Omidi, Tayfeh-Rezaie, Wassermann, J. Combin. Des. 15 (2007) 210–220, Tables II, III and IV), KT04 (Khosrovshahi, Tayfeh-Rezaie, lecture notes, 2004), Teirlinck84 (Ars Combin. 17 (1984) 173–176), Teirlinck89 (Discrete Math. 77 (1989) 345–356), and the semiregular families from ASL$(1,q)$ and AGL$(1,2^n)$ of the accompanying paper.

Non-existence: LS[5](2, 3, 7) (Cayley), LS[7](3, 4, 10) (Kramer and Mesner), LS[11](4, 5, 15) (Mendelsohn and Hung), LS[13](4, 5, 17) (Östergård and Pottonen).

Limitations

The tables contain only what follows from the sources above. Results published after 2010 are not included. For $N = 2$ no results beyond $k = 15$ for $t \geq 3$ are included. Large sets obtained by combining orbits of $\mathrm{PSL}(2, q)$ and $\mathrm{PGL}(2, q)$ are included for $q < 60$ from LOTW07 and for larger q only when the orbits have at most two lengths. One entry of Table II of LOTW07, $N = 29$ for $(3, 26, 60)$, is not admissible and was omitted. An entry marked open may therefore be settled in the literature.

Summary

Number of existing parameter sets over number of admissible ones.

N	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
2	1456/1456	1231/1287	1052/1155	878/1033	370/951
3	1299/1299	1164/1164	605/1039	166/928	28/827
5	631/922	304/756	26/640	2/580	0/522
7	859/916	350/721	48/602	6/517	0/462
11	656/656	69/512	0/384	0/276	0/184
13	286/769	52/629	6/501	3/389	1/289
17	304/798	61/693	0/594	0/504	0/420
19	488/834	87/714	0/600	0/495	0/396
23	244/878	81/778	0/682	0/592	0/506
29	905/905	124/827	1/751	0/679	0/609
31	268/847	107/763	0/681	0/603	0/527
37	475/856	16/805	0/755	0/707	0/660
41	257/890	74/833	0/777	0/723	0/670
43	287/862	60/802	0/743	0/686	0/630

Tables

LS[2](2, k, v)

All 1456 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

LS[2](3, k, v)

1231 of 1287 admissible parameter sets exist, 56 are open.

v	exists, k =	open, k =
11	4–5	
12	5–6	
13	6	
14	7	
19	4–9	
20	5–10	
21	6–10	
22	7–11	
23	8–11	
24	9–12	
25	10–12	
26	11–13	
27	4–7, 12–13	
28	5–7, 13–14	
29	6–7, 14	
30	7, 15	
35	4–15	16–17
36	5–15, 18	16–17
37	6–15	16–18
38	7–19	
39	8–19	

v	exists, k =	open, k =
40	9–20	
41	10–20	
42	11–21	
43	4–7, 12–21	
44	5–7, 13–22	
45	6–7, 14–22	
46	7, 15–23	
47	16–23	
48	17–24	
49	18–24	
50	19–25	
51	4–15, 20–25	
52	5–15, 21–26	
53	6–15, 22–26	
54	7–15, 23–27	
55	8–15, 24–27	
56	9–15, 25–28	
57	10–15, 26–28	
58	11–15, 27–29	
59	4–7, 12–15, 20–23, 28–29	
60	5–7, 13–15, 21–23, 29–30	
61	6–7, 14–15, 22–23, 30	
62	7, 15, 23, 31	
67	4–15, 20–31	16–19, 32–33
68	5–15, 19, 21–31, 34	16–18, 20, 32–33
69	6–15, 22–31	16–21, 32–34
70	7–31, 33, 35	32, 34
71	8–31	32–35
72	9–31, 33, 35–36	32, 34
73	10–31, 36	32–35
74	11–31, 33, 35, 37	32, 34, 36
75	4–7, 12–31, 36–37	32–35
76	5–7, 13–38	
77	6–7, 14–38	
78	7, 15–39	
79	16–39	
80	17–40	
81	18–40	
82	19–41	
83	4–15, 20–41	
84	5–15, 21–42	
85	6–15, 22–42	
86	7–15, 23–43	
87	8–15, 24–43	
88	9–15, 25–44	
89	10–15, 26–44	
90	11–15, 27–45	
91	4–7, 12–15, 20–23, 28–45	
92	5–7, 13–15, 21–23, 29–46	
93	6–7, 14–15, 22–23, 30–46	
94	7, 15, 23, 31–47	
95	32–47	
96	33–48	
97	34–48	
98	35–49	
99	4–15, 20–31, 36–49	16–19
100	5–15, 21–31, 37–50	16–20

$\mathrm{LS}[2](4, k, v)$

1052 of 1155 admissible parameter sets exist, 103 are open.

v	exists, k =	open, k =
12	5–6	
13	6	
14	7	
20	5–10	
21	6–10	
22	7–11	
23	8–11	
24	9–12	
25	10–12	

v	exists, k =	open, k =
26	11–13	
27	12–13	
28	5–7, 13–14	
29	6–7, 14	
30	7, 15	
36	5–15	16–18
37	6–15	16–18
38	7–15	16–19
39	8–15	16–19
40	9–20	
41	10–20	
42	11–21	
43	12–21	
44	5–7, 13–22	
45	6–7, 14–22	
46	7, 15–23	
47	16–23	
48	17–24	
49	18–24	
50	19–25	
51	20–25	
52	5–15, 21–26	
53	6–15, 22–26	
54	7–15, 23–27	
55	8–15, 24–27	
56	9–15, 25–28	
57	10–15, 26–28	
58	11–15, 27–29	
59	12–15, 28–29	
60	5–7, 13–15, 21–23, 29–30	
61	6–7, 14–15, 22–23, 30	
62	7, 15, 23, 31	
68	5–15, 21–31	16–20, 32–34
69	6–15, 22–31	16–21, 32–34
70	7–15, 23–31	16–22, 32–35
71	8–15, 24–31	16–23, 32–35
72	9–31	32–36
73	10–31	32–36
74	11–31, 37	32–36
75	12–31	32–37
76	5–7, 13–31, 37–38	32–36
77	6–7, 14–31, 38	32–37
78	7, 15–31, 33, 35, 37, 39	32, 34, 36, 38
79	16–31	32–39
80	17–40	
81	18–40	
82	19–41	
83	20–41	
84	5–15, 21–42	
85	6–15, 22–42	
86	7–15, 23–43	
87	8–15, 24–43	
88	9–15, 25–44	
89	10–15, 26–44	
90	11–15, 27–45	
91	12–15, 28–45	
92	5–7, 13–15, 21–23, 29–46	
93	6–7, 14–15, 22–23, 30–46	
94	7, 15, 23, 31–47	
95	32–47	
96	33–48	
97	34–48	
98	35–49	
99	36–49	
100	5–15, 21–31, 37–50	16–20

$\mathrm{LS}[2](5, k, v)$

878 of 1033 admissible parameter sets exist, 155 are open.

v	exists, k =	open, k =
13	6	

v	exists, k =	open, k =
14	7	
21	6–10	
22	7–11	
23	8–11	
24	9–12	
25	10–12	
26	11–13	
27	12–13	
28	13–14	
29	6–7, 14	
30	7, 15	
37	6–15	16–18
38	7–15	16–19
39	8–15	16–19
40	9–15	16–20
41	10–15	16–20
42	11–21	
43	12–21	
44	13–22	
45	6–7, 14–22	
46	7, 15–23	
47	16–23	
48	17–24	
49	18–24	
50	19–25	
51	20–25	
52	21–26	
53	6–15, 22–26	
54	7–15, 23–27	
55	8–15, 24–27	
56	9–15, 25–28	
57	10–15, 26–28	
58	11–15, 27–29	
59	12–15, 28–29	
60	13–15, 29–30	
61	6–7, 14–15, 22–23, 30	
62	7, 15, 23, 31	
69	6–15, 22–31	16–21, 32–34
70	7–15, 23–31	16–22, 32–35
71	8–15, 24–31	16–23, 32–35
72	9–15, 25–31	16–24, 32–36
73	10–15, 26–31	16–25, 32–36
74	11–31	32–37
75	12–31	32–37
76	13–31	32–38
77	6–7, 14–31, 38	32–37
78	7, 15–31, 39	32–38
79	16–31	32–39
80	17–31	32–40
81	18–31	32–40
82	19–31, 33, 35, 37, 39, 41	32, 34, 36, 38, 40
83	20–31	32–41
84	21–42	
85	6–15, 22–42	
86	7–15, 23–43	
87	8–15, 24–43	
88	9–15, 25–44	
89	10–15, 26–44	
90	11–15, 27–45	
91	12–15, 28–45	
92	13–15, 29–46	
93	6–7, 14–15, 22–23, 30–46	
94	7, 15, 23, 31–47	
95	32–47	
96	33–48	
97	34–48	
98	35–49	
99	36–49	
100	37–50	

$\mathrm{LS}[2](6, k, v)$

370 of 951 admissible parameter sets exist, 581 are open.

v	exists, k =	open, k =
14	7	
22	7–9, 11	10
23	8–9	10–11
24	9, 12	10–11
25		10–12
26	13	11–12
27		12–13
28	13–14	
29	14	
30	7, 15	
38	7–9, 15	10–14, 16–19
39	8–9	10–19
40	9	10–20
41		10–20
42		11–21
43		12–21
44	13–19, 21–22	20
45	14–19, 22	20–21
46	7, 15–19, 21, 23	20, 22
47	16–19	20–23
48	17–19, 21, 23–24	20, 22
49	18–19, 24	20–23
50	19, 21, 23, 25	20, 22, 24
51		20–25
52	21, 23, 25–26	22, 24
53	26	22–25
54	7–9, 15, 23, 25, 27	10–14, 24, 26
55	8–9	10–15, 24–27
56	9, 25–28	10–15
57	26–28	10–15
58	27–29	11–15
59	28–29	12–15
60	13–15, 29–30	
61	14–15, 30	
62	7, 15, 23, 31	
70	7–9, 15, 23, 31	10–14, 16–22, 24–30, 32–35
71	8–9	10–35
72	9	10–36
73		10–36
74		11–37
75		12–37
76	13–19, 21, 23, 25, 27, 29–31	20, 22, 24, 26, 28, 32–38
77	14–19, 30–31	20–29, 32–38
78	7, 15–19, 21, 23, 25, 27, 29, 31, 39	20, 22, 24, 26, 28, 30, 32–38
79	16–19	20–39
80	17–19, 21, 23, 25, 27, 29, 31	20, 22, 24, 26, 28, 30, 32–40
81	18–19	20–40
82	19, 21, 23, 25, 27, 29, 31	20, 22, 24, 26, 28, 30, 32–41
83		20–41
84	21, 23, 25, 27, 29, 31	22, 24, 26, 28, 30, 32–42
85		22–42
86	7–9, 15, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43	10–14, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42
87	8–9	10–15, 24–43
88	9, 25–39, 41–44	10–15, 40
89	26–39, 42–44	10–15, 40–41
90	27–39, 41, 43–45	11–15, 40, 42
91	28–39, 44–45	12–15, 40–43
92	13–15, 29–39, 41–43, 45–46	40, 44
93	14–15, 30–39, 42–43, 46	40–41, 44–45
94	7, 15, 23, 31–39, 41, 43, 45, 47	40, 42, 44, 46
95	32–39	40–47
96	33–39, 41–43, 45–48	40, 44
97	34–39, 42–43, 46–48	40–41, 44–45
98	35–39, 41, 43, 45, 47–49	40, 42, 44, 46
99	36–39, 48–49	40–47
100	37–39, 41–43, 45–47, 49–50	40, 44, 48

$\mathrm{LS}[3](2, k, v)$

All 1299 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

LS[3](3, k, v)

All 1164 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

LS[3](4, k, v)

605 of 1039 admissible parameter sets exist, 434 are open.

v	exists, k =	open, k =
13	5–6	
14	6–7	
15	7	
16	8	
22	5–8	
23	6–8	
24	7–8	
25	8	
31	5–10, 14–15	11–13
32	6–10, 15–16	11–14
33	7–10, 16	11–15
34	8–10, 17	11–16
35	9–10	11–17
36	10	11–18
37		11–18
38		12–19
39	13–19	
40	5–8, 14–20	
41	6–8, 15–20	
42	7–8, 16–21	
43	8, 17–21	
44	18–22	
45	19–22	
46	20–23	
47	21–23	
48	22–24	
49	5–8, 14–17, 23–24	
50	6–8, 15–17, 24–25	
51	7–8, 16–17, 25	
52	8, 17, 26	
58	5–10, 14–17, 23–26	11–13, 18–22
59	6–10, 15–17, 24–26	11–14, 18–23
60	7–10, 16–17, 25–26	11–15, 18–24
61	8–10, 17, 26	11–16, 18–25
62	9–10	11–26
63	10	11–26
64		11–26
65		12–26
66	13–26	
67	5–8, 14–26, 32–33	
68	6–8, 15–26, 33–34	
69	7–8, 16–26, 34	
70	8, 17–26, 35	
71	18–26	
72	19–26	
73	20–26	
74	21–26	
75	22–26	
76	5–8, 14–17, 23–26, 32–35	
77	6–8, 15–17, 24–26, 33–35	
78	7–8, 16–17, 25–26, 34–35	
79	8, 17, 26, 35	
85	5–10, 14–17, 23–26, 32–35, 41–42	11–13, 18–22, 27–31, 36–40
86	6–10, 15–17, 24–26, 33–35, 42–43	11–14, 18–23, 27–32, 36–41
87	7–10, 16–17, 25–26, 34–35, 43	11–15, 18–24, 27–33, 36–42
88	8–10, 17, 26, 35, 44	11–16, 18–25, 27–34, 36–43
89	9–10	11–44
90	10	11–45
91		11–45
92		12–46
93	13–32, 34–35, 37–38, 40–46	33, 36, 39
94	5–8, 14–35, 38, 41–47	36–37, 39–40
95	6–8, 15–35, 42–47	36–41
96	7–8, 16–35, 37–38, 40–41, 43–48	36, 39, 42
97	8, 17–35, 38, 41, 44–48	36–37, 39–40, 42–43

v	exists, $k =$	open, $k =$
98	18–35, 45–49	36–44
99	19–35, 37–38, 40–41, 43–44, 46–49	36, 39, 42, 45
100	20–35, 38, 41, 44, 47–50	36–37, 39–40, 42–43, 45–46

$\mathrm{LS}[3](5, k, v)$

166 of 928 admissible parameter sets exist, 762 are open.

v	exists, $k =$	open, $k =$
14	7	6
15		7
16	8	
23	6	7–8
24	7–8	
25	8	
32	9–10, 16	6–8, 11–15
33	7, 10	8–9, 11–16
34	17	8–16
35		9–17
36		10–18
37		11–18
38		12–19
39		13–19
40	20	14–19
41	6	7–8, 15–20
42	16–17, 19–21	7–8, 18
43	17, 20–21	8, 18–19
44	21–22	18–20
45	19–20, 22	21
46	20, 23	21–22
47		21–23
48	22–24	
49	23–24	
50	24–25	6–8, 15–17
51	25	7–8, 16–17
52	26	8, 17
59	6	7–26
60		7–26
61		8–26
62		9–26
63		10–26
64		11–26
65		12–26
66		13–26
67		14–26
68	34	6–8, 15–26, 33
69	16–20, 22–23, 25–26	7–8, 21, 24, 34
70	17–20, 23, 26, 35	8, 21–22, 24–25
71	18–20	21–26
72	19–26	
73	20–26	
74	21–26	
75	22–26	
76	23–26	
77	6, 24–26	7–8, 15–17, 33–35
78	25–26	7–8, 16–17, 34–35
79	26	8, 17, 35
86	43	6–42
87		7–43
88	44	8–43
89		9–44
90		10–45
91		11–45
92		12–46
93		13–46
94	47	14–46
95	6	7–8, 15–47
96	16–17, 19–20, 22–23, 25–32, 43–44, 46–48	7–8, 18, 21, 24, 33–42, 45
97	17, 20, 23, 26–32, 44, 47–48	8, 18–19, 21–22, 24–25, 33–43, 45–46
98	27–32, 48–49	18–26, 33–47
99	19–23, 25–26, 28–32, 46–47, 49	24, 27, 33–45, 48
100	20–23, 26, 29–32, 47, 50	24–25, 27–28, 33–46, 48–49

LS[3](6, k, v)

28 of 827 admissible parameter sets exist, 799 are open.

v	exists, k =	open, k =
15		7
16		8
24		7–8
25		8
33	10	7–9, 11–16
34		8–17
35		9–17
36		10–18
37		11–18
38		12–19
39		13–19
40		14–20
41		15–20
42		7–8, 16–21
43		8, 17–21
44		18–22
45	22	19–21
46	23	20–22
47		21–23
48		22–24
49		23–24
50		24–25
51	25	7–8, 16–17
52	26	8, 17
60		7–26
61		8–26
62		9–26
63		10–26
64		11–26
65		12–26
66		13–26
67		14–26
68		15–26
69		7–8, 16–26, 34
70		8, 17–26, 35
71		18–26
72	19–20	21–26
73	20	21–26
74		21–26
75	22–23, 25–26	24
76	23, 26	24–25
77		24–26
78	25–26	7–8, 16–17, 34–35
79	26	8, 17, 35
87		7–43
88		8–44
89		9–44
90		10–45
91		11–45
92		12–46
93		13–46
94		14–47
95		15–47
96		7–8, 16–48
97		8, 17–48
98		18–49
99	28–32, 49	19–27, 33–48
100	29–32, 50	20–28, 33–49

LS[5](2, k, v)

631 of 922 admissible parameter sets exist, 1 do not exist, 290 are open. Non-existent: (2, 3, 7).

v	exists, k =	open, k =
7		
8	4	
12	3–4	
13	4	

v	exists, k =	open, k =
17	3–4, 8	
18	4, 9	
22	3–4, 8–9	
23	4, 9	
27	3–8, 10–12	9, 13
28	4–8, 11–12	9–10, 13–14
29	5–8, 10–13	9, 14
30	6–8, 11–13	9–10, 14–15
31	7–14	15
32	3–4, 8–15	16
33	4, 9–15	16
34	10–15	16–17
35	11–17	
36	12–18	
37	3–4, 8, 13–18	9
38	4, 14–19	9
39	15–19	
40	16–20	
41	17–20	
42	3–4, 8, 18–21	9, 13–14
43	4, 19–21	9, 14
44	20–22	
45	21–22	
46	22–23	
47	3–4, 8–9, 13–14, 18–19, 23	
48	4, 9, 14, 19, 24	
52	3–8	9–24
53	4–11, 15–24	12–14
54	5–11, 16–24	12–15
55	6–11, 17–24	12–16
56	7–11, 18–24	12–17
57	3–4, 8–11, 19–24	12–18, 28
58	4, 9–11, 20–24	12–19, 29
59	10–24	
60	11–24	
61	12–24	
62	3–4, 8, 13–24	9, 28–29
63	4, 14–24	9, 29
64	15–24	
65	16–24	
66	17–24	
67	3–4, 8, 14, 18–24, 29	9, 13, 28, 33
68	4, 19–24	9, 14, 29, 34
69	20–24	
70	21–24	
71	22–24	
72	3–4, 8, 23–24	9, 13–14, 18–19, 28–29, 33–34
73	4, 14, 24	9, 19, 29, 34
77	3–8	9–24, 28–38
78	4–8	9–24, 29–39
79	5–8, 11, 17, 20, 23, 32, 35, 38	9–10, 12–16, 18–19, 21–22, 24, 30–31, 33–34, 36–37, 39
80	6–8	9–24, 31–40
81	7–8, 19, 22–23, 34, 38	9–18, 20–21, 24, 32–33, 35–37, 39–40
82	3–4, 8, 23	9–22, 24, 28–29, 33–41
83	4, 9–24, 29, 34–40	41
84	10–24, 35–40	41–42
85	11–24, 36–42	
86	12–24, 37–43	
87	3–4, 8, 13–24, 38–43	9, 28–29, 33–34
88	4, 14–24, 39–44	9, 29, 34
89	15–24, 40–44	
90	16–24, 41–45	
91	17–24, 42–45	
92	3–4, 8, 18–24, 43–46	9, 13–14, 28–29, 33–34, 38–39
93	4, 19–24, 44–46	9, 14, 29, 34, 39
94	20–24, 45–47	
95	21–24, 46–47	
96	22–24, 47–48	
97	3–4, 8, 23–24, 38, 48	9, 13–14, 18–19, 28–29, 33–34, 39, 43–44
98	4, 24, 49	9, 14, 19, 29, 34, 39, 44

LS[5](3, k, v)

304 of 756 admissible parameter sets exist, 1 do not exist, 451 are open. Non-existent: (3, 4, 8).

v	exists, k =	open, k =
8		
13	4	
18	4, 9	
23	4, 9	
28	4–6, 8, 11–12	7, 9–10, 13–14
29	5–6, 12	7–11, 13–14
30	6, 11–13	7–10, 14–15
31	12–13	7–11, 14–15
32	9, 11, 13–14	8, 10, 12, 15–16
33	4, 9, 14	10–13, 15–16
34		10–17
35		11–17
36	18	12–17
37		13–18
38	4, 14–19	9
39	15–19	
40	16–20	
41	17–20	
42	18–21	
43	4, 19–21	9, 14
44	20–22	
45	21–22	
46	22–23	
47	23	
48	4, 9, 14, 19, 24	
53	4–6	7–24
54	5–8, 10–11, 16–17, 19–24	9, 12–15, 18
55	6–8, 11, 17, 20–24	9–10, 12–16, 18–19
56	7–8, 21–24	9–20
57	8, 22–24	9–21
58	4, 23–24	9–22, 29
59	24	10–23
60	11–23	24
61	12–23	24
62	13–23	24
63	4, 14–23	9, 24, 29
64	15–23	24
65	16–24	
66	17–24	
67	18–24	
68	4, 19–24	9, 14, 29, 34
69	20–24	
70	21–24	
71	22–24	
72	23–24	
73	4, 24	9, 14, 19, 29, 34
78	4–6	7–24, 29–39
79	5–6	7–24, 30–39
80	6	7–24, 31–40
81		7–24, 32–40
82		8–24, 33–41
83	4	9–24, 29, 34–41
84	11, 13, 15, 17, 19, 23, 37, 39	10, 12, 14, 16, 18, 20–22, 24, 35–36, 38, 40–42
85		11–24, 36–42
86	43	12–24, 37–42
87		13–24, 38–43
88	4, 44	9, 14–24, 29, 34, 39–43
89		15–24, 40–44
90	16–24, 41–45	
91	17–24, 42–45	
92	18–24, 43–46	
93	4, 19–24, 44–46	9, 14, 29, 34, 39
94	20–24, 45–47	
95	21–24, 46–47	
96	22–24, 47–48	
97	23–24, 48	
98	4, 24, 49	9, 14, 19, 29, 34, 39, 44

LS[5](4, k, v)

26 of 640 admissible parameter sets exist, 614 are open.

v	exists, k =	open, k =
29		5–14
30		6–15
31		7–15
32		8–16
33		9–16
34		10–17
35		11–17
36		12–18
37		13–18
38		14–19
39		15–19
40		16–20
41		17–20
42		18–21
43		19–21
44		20–22
45		21–22
46		22–23
47		23
48		24
54		5–24
55		6–24
56		7–24
57		8–24
58		9–24
59		10–24
60		11–24
61		12–24
62		13–24
63		14–24
64		15–24
65		16–24
66		17–24
67		18–24
68		19–24
69		20–24
70	21–24	
71	22–24	
72	23–24	
73	24	
79		5–24, 30–39
80		6–24, 31–40
81		7–24, 32–40
82		8–24, 33–41
83		9–24, 34–41
84		10–24, 35–42
85		11–24, 36–42
86		12–24, 37–43
87		13–24, 38–43
88		14–24, 39–44
89		15–24, 40–44
90		16–24, 41–45
91		17–24, 42–45
92		18–24, 43–46
93		19–24, 44–46
94		20–24, 45–47
95	21–24, 46–47	
96	22–24, 47–48	
97	23–24, 48	
98	24, 49	

LS[5](5, k, v)

2 of 580 admissible parameter sets exist, 578 are open.

v	exists, k =	open, k =
30		6–15
31		7–15

v	exists, k =	open, k =
32		8–16
33		9–16
34		10–17
35		11–17
36		12–18
37		13–18
38		14–19
39		15–19
40		16–20
41		17–20
42		18–21
43		19–21
44		20–22
45		21–22
46		22–23
47		23
48		24
55		6–24
56		7–24
57		8–24
58		9–24
59		10–24
60		11–24
61		12–24
62		13–24
63		14–24
64		15–24
65		16–24
66		17–24
67		18–24
68		19–24
69		20–24
70		21–24
71		22–24
72		23–24
73		24
80		6–24, 31–40
81		7–24, 32–40
82		8–24, 33–41
83		9–24, 34–41
84		10–24, 35–42
85		11–24, 36–42
86		12–24, 37–43
87		13–24, 38–43
88		14–24, 39–44
89		15–24, 40–44
90		16–24, 41–45
91		17–24, 42–45
92		18–24, 43–46
93		19–24, 44–46
94		20–24, 45–47
95		21–24, 46–47
96	48	22–24, 47
97		23–24, 48
98	49	24

$\mathrm{LS}[5](6, k, v)$

None of the 522 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

$\mathrm{LS}[7](2, k, v)$

859 of 916 admissible parameter sets exist, 57 are open.

v	exists, k =	open, k =
9	3–4	
10	4–5	
11	5	
12	6	
16	3–6	
17	4–6	

v	exists, k =	open, k =
18	5–6	
19	6	
23	3–6, 10–11	
24	4–6, 11–12	
25	5–6, 12	
26	6, 13	
30	3–6, 10–13	
31	4–6, 11–13	
32	5–6, 12–13	
33	6, 13	
37	3–6, 10–13, 17–18	
38	4–6, 11–13, 18–19	
39	5–6, 12–13, 19	
40	6, 13, 20	
44	3–6, 10–13, 17–20	
45	4–6, 11–13, 18–20	
46	5–6, 12–13, 19–20	
47	6, 13, 20	
51	3–6, 10–13, 17–20, 24–25	7–9, 14–16, 21–23
52	4–6, 11–13, 18–20, 25–26	7–10, 14–17, 21–24
53	5–13, 15–26	14
54	6–13, 16–27	14–15
55	7–13, 17–27	14–16
56	8–13, 18–28	14–17
57	9–13, 19–28	14–18
58	3–6, 10–13, 17–29	14–16
59	4–6, 11–29	
60	5–6, 12–30	
61	6, 13–30	
62	14–31	
63	15–31	
64	16–32	
65	3–6, 10–13, 17–32	
66	4–6, 11–13, 18–33	
67	5–6, 12–13, 19–33	
68	6, 13, 20–34	
69	21–34	
70	22–35	
71	23–35	
72	3–6, 10–13, 17–20, 24–36	
73	4–6, 11–13, 18–20, 25–36	
74	5–6, 12–13, 19–20, 26–37	
75	6, 13, 20, 27–37	
76	28–38	
77	29–38	
78	30–39	
79	3–6, 10–13, 17–20, 24–27, 31–39	
80	4–6, 11–13, 18–20, 25–27, 32–40	
81	5–6, 12–13, 19–20, 26–27, 33–40	
82	6, 13, 20, 27, 34–41	
83	35–41	
84	36–42	
85	37–42	
86	3–6, 10–13, 17–20, 24–27, 31–34, 38–43	
87	4–6, 11–13, 18–20, 25–27, 32–34, 39–43	
88	5–6, 12–13, 19–20, 26–27, 33–34, 40–44	
89	6, 13, 20, 27, 34, 41–44	
90	42–45	
91	43–45	
92	44–46	
93	3–6, 10–13, 17–20, 24–27, 31–34, 38–41, 45–46	
94	4–6, 11–13, 18–20, 25–27, 32–34, 39–41, 46–47	
95	5–6, 12–13, 19–20, 26–27, 33–34, 40–41, 47	
96	6, 13, 20, 27, 34, 41, 48	
100	3–6, 10–13, 17–20, 24–27, 31–34, 38–41, 45–48	7–9, 14–16, 21–23, 28–30, 35–37, 42–44

$\mathrm{LS}[7](3, k, v)$

350 of 721 admissible parameter sets exist, 1 do not exist, 370 are open. Non-existent: $(3, 4, 10)$.

v	exists, k =	open, k =
10	5	

v	exists, k =	open, k =
11	5	
12	6	
17	4–6	
18	5–6	
19	6	
24	4–6, 12	11
25	5–6	12
26	6, 13	
31	4–6	11–13
32	5–6, 12–13	
33	6, 13	
38	4–6, 12–13, 18–19	11
39	5–6, 13, 19	12
40	6, 20	13
45	4–6	11–13, 18–20
46	5–6	12–13, 19–20
47	6	13, 20
52	4–6, 26	7–25
53	5–6	7–26
54	6–8, 10–11, 16–17, 19–25, 27	9, 12–15, 18, 26
55	7–8, 11, 17, 20–25	9–10, 12–16, 18–19, 26–27
56	8, 21–25, 28	9–20, 26–27
57	22–25	9–21, 26–28
58	23–25, 29	10–22, 26–28
59	4–6, 24–25	11–23, 26–29
60	5–6, 12–17, 19–23, 25–28, 30	18, 24, 29
61	6, 13–17, 20–23, 26–28	18–19, 24–25, 29–30
62	14–17, 21–23, 27–28, 31	18–20, 24–26, 29–30
63	15–17, 22–23, 28	18–21, 24–27, 29–31
64	16–17, 23, 32	18–22, 24–31
65	17	18–32
66	4–6, 33	11–13, 18–32
67	5–6	12–13, 19–33
68	6, 21, 25, 34	13, 20, 22–24, 26–33
69		21–34
70	22–27, 29–35	28
71	23–27, 30–35	28–29
72	24–27, 31–36	28–30
73	4–6, 25–27, 32–36	11–13, 18–20, 28–31
74	5–6, 26–27, 33–37	12–13, 19–20, 28–32
75	6, 27, 34–37	13, 20, 28–33
76	35–38	28–34
77	29–38	
78	30–39	
79	31–39	
80	4–6, 32–40	11–13, 18–20, 25–27
81	5–6, 33–40	12–13, 19–20, 26–27
82	6, 34–41	13, 20, 27
83	35–41	
84	36–42	
85	37–42	
86	38–43	
87	4–6, 39–43	11–13, 18–20, 25–27, 32–34
88	5–6, 40–44	12–13, 19–20, 26–27, 33–34
89	6, 41–44	13, 20, 27, 34
90	42–45	
91	43–45	
92	44–46	
93	45–46	
94	4–6, 46–47	11–13, 18–20, 25–27, 32–34, 39–41
95	5–6, 47	12–13, 19–20, 26–27, 33–34, 40–41
96	6, 48	13, 20, 27, 34, 41

$\mathrm{LS}[7](4, k, v)$

48 of 602 admissible parameter sets exist, 1 do not exist, 553 are open. Non-existent: $(4, 5, 11)$.

v	exists, k =	open, k =
11		
12		6
18		5–6
19		6

v	exists, $k =$	open, $k =$
25		5–6, 12
26		6, 13
32	5	6, 12–13
33		6, 13
39		5–6, 12–13, 19
40		6, 13, 20
46		5–6, 12–13, 19–20
47		6, 13, 20
53		5–26
54		6–27
55		7–27
56		8–28
57		9–28
58		10–29
59		11–29
60	5	6, 12–30
61		6, 13–30
62		14–31
63		15–31
64		16–32
65		17–32
66		18–33
67		5–6, 12–13, 19–33
68		6, 13, 20–34
69		21–34
70		22–35
71		23–35
72		24–36
73		25–36
74		5–6, 12–13, 19–20, 26–37
75		6, 13, 20, 27–37
76		28–38
77	36–38	29–35
78	37–39	30–36
79	38–39	31–37
80	39–40	32–38
81	40	5–6, 12–13, 19–20, 26–27, 33–39
82	41	6, 13, 20, 27, 34–40
83		35–41
84	36–41	42
85	37–41	42
86	38–41	42–43
87	39–41	42–43
88	5, 40–41	6, 12–13, 19–20, 26–27, 33–34, 42–44
89	41	6, 13, 20, 27, 34, 42–44
90		42–45
91	43–45	
92	44–46	
93	45–46	
94	46–47	
95	47	5–6, 12–13, 19–20, 26–27, 33–34, 40–41
96	48	6, 13, 20, 27, 34, 41

$\mathrm{LS}[7](5, k, v)$

6 of 517 admissible parameter sets exist, 1 do not exist, 510 are open. Non-existent: $(5, 6, 12)$.

v	exists, $k =$	open, $k =$
12		
19		6
26		6, 13
33		6, 13
40		6, 13, 20
47		6, 13, 20
54		6–27
55		7–27
56		8–28
57		9–28
58		10–29
59		11–29
60		12–30
61		6, 13–30

v	exists, $k =$	open, $k =$
62		14–31
63		15–31
64		16–32
65		17–32
66		18–33
67		19–33
68		6, 13, 20–34
69		21–34
70		22–35
71		23–35
72		24–36
73		25–36
74		26–37
75		6, 13, 20, 27–37
76		28–38
77		29–38
78	39	30–38
79		31–39
80	40	32–39
81		33–40
82	41	6, 13, 20, 27, 34–40
83		35–41
84		36–42
85		37–42
86		38–43
87		39–43
88		40–44
89		6, 13, 20, 27, 34, 41–44
90		42–45
91		43–45
92	46	44–45
93		45–46
94	47	46
95		47
96	48	6, 13, 20, 27, 34, 41

$\mathrm{LS}[7](6, k, v)$

None of the 462 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

$\mathrm{LS}[11](2, k, v)$

All 656 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

$\mathrm{LS}[11](3, k, v)$

69 of 512 admissible parameter sets exist, 443 are open.

v	exists, $k =$	open, $k =$
14	7	4–6
15		5–7
16	8	6–7
17		7–8
18	9	8
19		9
20	10	
25		4–10
26		5–10
27		6–10
28		7–10
29		8–10
30		9–10
31		10
36	4, 18	5–10, 15–17
37		5–10, 16–18
38	17–19	6–10
39	18–19	7–10
40	19–20	8–10
41	20	9–10
42	21	10

v	exists, k =	open, k =
47	**4**	5–10, 15–21
48	5–7, 9–10, 17–21	8, 16
49	6–7, 10, 18–21	8–9, 17
50	7, 19–21	8–10, 18
51	20–21	8–10, 19
52	21	9–10, 20
53		10, 21
58	29	4–10, 15–21, 26–28
59		5–10, 16–21, 27–29
60	7–9, 17, 19–21, 30	6, 10, 18, 28–29
61	8–9, 20–21	7, 10, 18–19, 29–30
62	9, 21, 31	8, 10, 19–20, 30
63		9–10, 20–21, 31
64	32	10, 21
69	4	5–10, 15–21, 26–32
70		5–10, 16–21, 27–32
71		6–10, 17–21, 28–32
72	19, 29	7–10, 18, 20–21, 30–32
73		8–10, 19–21, 30–32
74		9–10, 20–21, 31–32
75		10, 21, 32
80	**4**, 40	5–10, 15–21, 26–32, 37–39
81		5–10, 16–21, 27–32, 38–40
82	41	6–10, 17–21, 28–32, 39–40
83		7–10, 18–21, 29–32, 40–41
84	9, 19, 31, 42	8, 10, 20–21, 30, 32, 41
85		9–10, 20–21, 31–32, 42
86	43	10, 21, 32
91	**4**	5–10, 15–21, 26–32, 37–43
92		5–10, 16–21, 27–32, 38–43
93		6–10, 17–21, 28–32, 39–43
94		7–10, 18–21, 29–32, 40–43
95		8–10, 19–21, 30–32, 41–43
96		9–10, 20–21, 31–32, 42–43
97		10, 21, 32, 43

$\mathrm{LS}[11](4, k, v)$

0 of 384 admissible parameter sets exist, 1 do not exist, 383 are open. Non-existent: $(4, 5, 15)$.

v	exists, k =	open, k =
15		6–7
16		6–8
17		7–8
18		8–9
19		9
20		10
26		5–10
27		6–10
28		7–10
29		8–10
30		9–10
31		10
37		5–10, 16–18
38		6–10, 17–19
39		7–10, 18–19
40		8–10, 19–20
41		9–10, 20
42		10, 21
48		5–10, 16–21
49		6–10, 17–21
50		7–10, 18–21
51		8–10, 19–21
52		9–10, 20–21
53		10, 21
59		5–10, 16–21, 27–29
60		6–10, 17–21, 28–30
61		7–10, 18–21, 29–30
62		8–10, 19–21, 30–31
63		9–10, 20–21, 31
64		10, 21, 32
70		5–10, 16–21, 27–32

v	exists, $k =$	open, $k =$
71		6–10, 17–21, 28–32
72		7–10, 18–21, 29–32
73		8–10, 19–21, 30–32
74		9–10, 20–21, 31–32
75		10, 21, 32
81		5–10, 16–21, 27–32, 38–40
82		6–10, 17–21, 28–32, 39–41
83		7–10, 18–21, 29–32, 40–41
84		8–10, 19–21, 30–32, 41–42
85		9–10, 20–21, 31–32, 42
86		10, 21, 32, 43
92		5–10, 16–21, 27–32, 38–43
93		6–10, 17–21, 28–32, 39–43
94		7–10, 18–21, 29–32, 40–43
95		8–10, 19–21, 30–32, 41–43
96		9–10, 20–21, 31–32, 42–43
97		10, 21, 32, 43

$\mathrm{LS}[11](5, k, v)$

0 of 276 admissible parameter sets exist, 1 do not exist, 275 are open. Non-existent: $(5, 6, 16)$.

v	exists, $k =$	open, $k =$
16		7–8
17		7–8
18		8–9
19		9
20		10
27		6–10
28		7–10
29		8–10
30		9–10
31		10
38		6–10, 17–19
39		7–10, 18–19
40		8–10, 19–20
41		9–10, 20
42		10, 21
49		6–10, 17–21
50		7–10, 18–21
51		8–10, 19–21
52		9–10, 20–21
53		10, 21
60		6–10, 17–21, 28–30
61		7–10, 18–21, 29–30
62		8–10, 19–21, 30–31
63		9–10, 20–21, 31
64		10, 21, 32
71		6–10, 17–21, 28–32
72		7–10, 18–21, 29–32
73		8–10, 19–21, 30–32
74		9–10, 20–21, 31–32
75		10, 21, 32
82		6–10, 17–21, 28–32, 39–41
83		7–10, 18–21, 29–32, 40–41
84		8–10, 19–21, 30–32, 41–42
85		9–10, 20–21, 31–32, 42
86		10, 21, 32, 43
93		6–10, 17–21, 28–32, 39–43
94		7–10, 18–21, 29–32, 40–43
95		8–10, 19–21, 30–32, 41–43
96		9–10, 20–21, 31–32, 42–43
97		10, 21, 32, 43

$\mathrm{LS}[11](6, k, v)$

0 of 184 admissible parameter sets exist, 1 do not exist, 183 are open. Non-existent: $(6, 7, 17)$.

v	exists, k =	open, k =
17		8
18		8–9
19		9
20		10
28		7–10
29		8–10
30		9–10
31		10
39		7–10, 18–19
40		8–10, 19–20
41		9–10, 20
42		10, 21
50		7–10, 18–21
51		8–10, 19–21
52		9–10, 20–21
53		10, 21
61		7–10, 18–21, 29–30
62		8–10, 19–21, 30–31
63		9–10, 20–21, 31
64		10, 21, 32
72		7–10, 18–21, 29–32
73		8–10, 19–21, 30–32
74		9–10, 20–21, 31–32
75		10, 21, 32
83		7–10, 18–21, 29–32, 40–41
84		8–10, 19–21, 30–32, 41–42
85		9–10, 20–21, 31–32, 42
86		10, 21, 32, 43
94		7–10, 18–21, 29–32, 40–43
95		8–10, 19–21, 30–32, 41–43
96		9–10, 20–21, 31–32, 42–43
97		10, 21, 32, 43

$\mathrm{LS}[13](2, k, v)$

286 of 769 admissible parameter sets exist, 483 are open.

v	exists, k =	open, k =
15	3–4, 7	5–6
16	4, 8	5–7
17		5–8
18		6–9
19	8	7, 9
20		8–10
21		9–10
22		10–11
23		11
24		12
28	3–4	5–12
29	4, 6–11	5, 12
30	7–11	5–6, 12
31	7–12	6
32	7–12	
33	8–12	
34	9–12	
35	10–12	
36	11–12	
37	12	
41	3–4, 18–19	5–12, 16–17, 20
42	4, 19	5–12, 17–18, 20–21
43	5, 9–12, 18–20	6–8, 21
44	10–12, 19–20	6–9, 21–22
45	11–12, 20	7–10, 21–22
46	12	8–11, 21–23
47	9–12, 22	23
48	10–12	23–24
49	11–12	24
50	12	25
54	3–4	5–12, 16–25
55	4	5–12, 17–25
56		5–12, 18–25
57		6–12, 19–25

v	exists, $k =$	open, $k =$
58		7–12, 20–25
59	8–12, 21–25	
60	9–12, 22–25	
61	10–12, 23–25	
62	11–12, 24–25	
63	12, 25	
67	3–8, 16–21, 24–25, 29, 32	9–12, 22–23, 30–31, 33
68	4–8, 17–21, 25	9–12, 22–24, 30–34
69	5–8, 18–21	9–12, 22–25, 31–34
70	6–8, 19–21	9–12, 22–25, 32–35
71	7–9, 12, 20–21, 23–24, 33–34	10–11, 22, 25, 35
72	8–9, 21, 24, 34	10–12, 22–23, 25, 35–36
73	9, 11, 23, 35	10, 12, 22, 24–25, 36
74		10–12, 23–25, 36–37
75		11–12, 24–25, 37
76		12, 25, 38
80	3–4, 7–8	5–6, 9–12, 16–25, 29–38
81	4, 8, 19, 22–23, 34, 38	5–7, 9–12, 17–18, 20–21, 24–25, 30–33, 35–37
82	23	5–12, 18–22, 24–25, 31–38
83	6–12, 19–25, 32–38	
84	7–12, 20–25, 33–38	
85	8–12, 21–25, 34–38	
86	9–12, 22–25, 35–38	
87	10–12, 23–25, 36–38	
88	11–12, 24–25, 37–38	
89	12, 25, 38	
93	3–4	5–12, 16–25, 29–38, 42–46
94	4, 6–8	5, 9–12, 17–25, 30–38, 43–47
95	7–8	5–6, 9–12, 18–25, 31–38, 44–47
96	7–8	6, 9–12, 19–25, 32–38, 45–48
97	7–8, 35, 38, 47	9–12, 20–25, 33–34, 36–37, 46, 48
98	8	9–12, 21–25, 34–38, 47–49
99		9–12, 22–25, 35–38, 48–49
100		10–12, 23–25, 36–38, 49–50

$\mathrm{LS}[13](3, k, v)$

52 of 629 admissible parameter sets exist, 577 are open.

v	exists, $k =$	open, $k =$
16	8	4–7
17		5–8
18		6–9
19		7–9
20		8–10
21		9–10
22		10–11
23		11
24		12
29		4–12
30	7–11	5–6, 12
31	8–11	6–7, 12
32	9–12	7–8
33	10–12	8–9
34	11–12	9–10
35	12	10–11
36		11–12
37		12
42	4, 19	5–12, 17–18, 20–21
43		5–12, 18–21
44	10, 12, 19–20	6–9, 11, 21–22
45	20	7–12, 21–22
46		8–12, 21–23
47		9–12, 22–23
48	10–12	23–24
49	11–12	24
50	12	25
55		4–12, 17–25
56		5–12, 18–25
57		6–12, 19–25
58		7–12, 20–25
59		8–12, 21–25

v	exists, k =	open, k =
60	9, 11–12, 22–23, 25	10, 24
61	12, 23	10–11, 24–25
62		11–12, 24–25
63		12, 25
68	7, 19, 21, 25	4–6, 8–12, 17–18, 20, 22–24, 30–34
69		5–12, 18–25, 31–34
70		6–12, 19–25, 32–35
71		7–12, 20–25, 33–35
72		8–12, 21–25, 34–36
73		9–12, 22–25, 35–36
74		10–12, 23–25, 36–37
75		11–12, 24–25, 37
76		12, 25, 38
81	4	5–12, 17–25, 30–38
82		5–12, 18–25, 31–38
83		6–12, 19–25, 32–38
84	9, 11, 23, 25, 33, 37	7–8, 10, 12, 20–22, 24, 34–36, 38
85		8–12, 21–25, 34–38
86		9–12, 22–25, 35–38
87		10–12, 23–25, 36–38
88		11–12, 24–25, 37–38
89		12, 25, 38
94		4–12, 17–25, 30–38, 43–47
95		5–12, 18–25, 31–38, 44–47
96		6–12, 19–25, 32–38, 45–48
97		7–12, 20–25, 33–38, 46–48
98		8–12, 21–25, 34–38, 47–49
99		9–12, 22–25, 35–38, 48–49
100		10–12, 23–25, 36–38, 49–50

$\mathrm{LS}[13](4, k, v)$

6 of 501 admissible parameter sets exist, 1 do not exist, 494 are open. Non-existent: $(4, 5, 17)$.

v	exists, k =	open, k =
17		6–8
18		6–9
19		7–9
20		8–10
21		9–10
22		10–11
23		11
24		12
30		5–12
31	8–10	6–7, 11–12
32	9–10	7–8, 11–12
33	10	8–9, 11–12
34		9–12
35		10–12
36		11–12
37		12
43		5–12, 18–21
44		6–12, 19–22
45		7–12, 20–22
46		8–12, 21–23
47		9–12, 22–23
48		10–12, 23–24
49		11–12, 24
50		12, 25
56		5–12, 18–25
57		6–12, 19–25
58		7–12, 20–25
59		8–12, 21–25
60		9–12, 22–25
61		10–12, 23–25
62		11–12, 24–25
63		12, 25
69		5–12, 18–25, 31–34
70		6–12, 19–25, 32–35
71		7–12, 20–25, 33–35
72		8–12, 21–25, 34–36
73		9–12, 22–25, 35–36

v	exists, $k =$	open, $k =$
74		10–12, 23–25, 36–37
75		11–12, 24–25, 37
76		12, 25, 38
82		5–12, 18–25, 31–38
83		6–12, 19–25, 32–38
84		7–12, 20–25, 33–38
85		8–12, 21–25, 34–38
86		9–12, 22–25, 35–38
87		10–12, 23–25, 36–38
88		11–12, 24–25, 37–38
89		12, 25, 38
95		5–12, 18–25, 31–38, 44–47
96		6–12, 19–25, 32–38, 45–48
97		7–12, 20–25, 33–38, 46–48
98		8–12, 21–25, 34–38, 47–49
99		9–12, 22–25, 35–38, 48–49
100		10–12, 23–25, 36–38, 49–50

$\mathrm{LS}[13](5, k, v)$

3 of 389 admissible parameter sets exist, 1 do not exist, 385 are open. Non-existent: $(5, 6, 18)$.

v	exists, $k =$	open, $k =$
18		7–9
19		7–9
20		8–10
21		9–10
22		10–11
23		11
24		12
31		6–12
32	9–10	7–8, 11–12
33	10	8–9, 11–12
34		9–12
35		10–12
36		11–12
37		12
44		6–12, 19–22
45		7–12, 20–22
46		8–12, 21–23
47		9–12, 22–23
48		10–12, 23–24
49		11–12, 24
50		12, 25
57		6–12, 19–25
58		7–12, 20–25
59		8–12, 21–25
60		9–12, 22–25
61		10–12, 23–25
62		11–12, 24–25
63		12, 25
70		6–12, 19–25, 32–35
71		7–12, 20–25, 33–35
72		8–12, 21–25, 34–36
73		9–12, 22–25, 35–36
74		10–12, 23–25, 36–37
75		11–12, 24–25, 37
76		12, 25, 38
83		6–12, 19–25, 32–38
84		7–12, 20–25, 33–38
85		8–12, 21–25, 34–38
86		9–12, 22–25, 35–38
87		10–12, 23–25, 36–38
88		11–12, 24–25, 37–38
89		12, 25, 38
96		6–12, 19–25, 32–38, 45–48
97		7–12, 20–25, 33–38, 46–48
98		8–12, 21–25, 34–38, 47–49
99		9–12, 22–25, 35–38, 48–49
100		10–12, 23–25, 36–38, 49–50

LS[13](6, k, v)

1 of 289 admissible parameter sets exist, 1 do not exist, 287 are open. Non-existent: (6, 7, 19).

v	exists, k =	open, k =
19		8–9
20		8–10
21		9–10
22		10–11
23		11
24		12
32		7–12
33	10	8–9, 11–12
34		9–12
35		10–12
36		11–12
37		12
45		7–12, 20–22
46		8–12, 21–23
47		9–12, 22–23
48		10–12, 23–24
49		11–12, 24
50		12, 25
58		7–12, 20–25
59		8–12, 21–25
60		9–12, 22–25
61		10–12, 23–25
62		11–12, 24–25
63		12, 25
71		7–12, 20–25, 33–35
72		8–12, 21–25, 34–36
73		9–12, 22–25, 35–36
74		10–12, 23–25, 36–37
75		11–12, 24–25, 37
76		12, 25, 38
84		7–12, 20–25, 33–38
85		8–12, 21–25, 34–38
86		9–12, 22–25, 35–38
87		10–12, 23–25, 36–38
88		11–12, 24–25, 37–38
89		12, 25, 38
97		7–12, 20–25, 33–38, 46–48
98		8–12, 21–25, 34–38, 47–49
99		9–12, 22–25, 35–38, 48–49
100		10–12, 23–25, 36–38, 49–50

LS[17](2, k, v)

304 of 798 admissible parameter sets exist, 494 are open.

v	exists, k =	open, k =
19	3–8	9
20	4–8	9–10
21	5–8	9–10
22	6–8	9–11
23	7–10	11
24	8–10	11–12
25	9–11	12
26	10–11	12–13
27	11	12–13
28		12–14
29		13–14
30		14–15
31	15	
32	16	
36	3–8	9–16
37	4–8, 11–16	9–10
38	5–8, 12–16	9–11
39	6–8, 13–16	9–12
40	7–8, 14–16	9–13
41	8, 12–16	9–11
42	13–16	9–12
43	11–16	10

v	exists, k =	open, k =
44	12–16	11
45	13–16	12
46	14–16	13
47	14–16	
48	15–16	
49	16	
53	3–11, 15, 20–25	12–14, 16, 26
54	4–11, 21–25	12–16, 26–27
55	5–11, 22–25	12–16, 26–27
56	6–11, 23–25	12–16, 26–28
57	7–11, 24–25	12–16, 26–28
58	8–11, 25	12–16, 26–29
59	9–16, 26–28	29
60	10–16, 27–28	29–30
61	11–16, 28	29–30
62	12–16	29–31
63	13–16	30–31
64	14–16	31–32
65	15–16	32
66	16	33
70	3–8, 11	9–10, 12–16, 20–33
71	4–9, 12–13, 23–24, 27–29, 32–33	10–11, 14–16, 21–22, 25–26, 30–31
72	5–9, 13, 24, 28–29, 33	10–12, 14–16, 22–23, 25–27, 30–32
73	6–9, 11, 14, 23, 26, 29	10, 12–13, 15–16, 24–25, 27–28, 30–33
74	7–10	11–16, 24–33
75	8–10	11–16, 25–33
76	9–11	12–16, 26–33
77	10–11	12–16, 27–33
78	11	12–16, 28–33
79	29, 32	12–16, 30–31, 33
80		13–16, 30–33
81		14–16, 31–33
82		15–16, 32–33
83	16, 33	
87	3–8	9–16, 20–33, 37–43
88	4–8, 11	9–10, 12–16, 21–33, 38–44
89	5–8, 10, 14–15, 26–27, 30–31, 39, 42–43	9, 11–13, 16, 22–25, 28–29, 32–33, 40–41, 44
90	6–8, 15, 27, 31, 43	9–14, 16, 23–26, 28–30, 32–33, 40–42, 44–45
91	7–8	9–16, 24–33, 41–45
92	8	9–16, 25–33, 42–46
93		9–16, 26–33, 43–46
94	11	10, 12–16, 27–33, 44–47
95		11–16, 28–33, 45–47
96		12–16, 29–33, 46–48
97	47	13–16, 30–33, 48
98		14–16, 31–33, 48–49
99		15–16, 32–33, 49
100		16, 33, 50

$\mathrm{LS}[17](3, k, v)$

61 of 693 admissible parameter sets exist, 632 are open.

v	exists, k =	open, k =
20		4–10
21		5–10
22		6–11
23		7–11
24	10	8–9, 11–12
25		9–12
26		10–13
27		11–13
28		12–14
29		13–14
30		14–15
31		15
32	16	
37		4–16
38	12–16	5–11
39	13–16	6–12
40	14–16	7–13
41	15–16	8–14

v	exists, k =	open, k =
42	13, 16	9–12, 14–15
43		10–16
44	12–13	11, 14–16
45	13	12, 14–16
46		13–16
47		14–16
48	15	16
49		16
54	4, 7–8, 10–11, 21, 23–25	5–6, 9, 12–16, 22, 26–27
55	8, 11, 24–25	5–7, 9–10, 12–16, 22–23, 26–27
56	25	6–16, 23–24, 26–28
57		7–16, 24–28
58		8–16, 25–29
59		9–16, 26–29
60	11–16, 27	10, 28–30
61	12–16	11, 28–30
62	13–16	12, 29–31
63	14–16	13, 30–31
64	15–16	14, 31–32
65	16	15, 32
66		16, 33
71	**4**	5–16, 21–33
72	13, 29	5–12, 14–16, 22–28, 30–33
73		6–16, 23–33
74		7–16, 24–33
75		8–16, 25–33
76		9–16, 26–33
77		10–16, 27–33
78		11–16, 28–33
79		12–16, 29–33
80		13–16, 30–33
81		14–16, 31–33
82		15–16, 32–33
83		16, 33
88		4–16, 21–33, 38–44
89		5–16, 22–33, 39–44
90		6–16, 23–33, 40–45
91		7–16, 24–33, 41–45
92		8–16, 25–33, 42–46
93		9–16, 26–33, 43–46
94		10–16, 27–33, 44–47
95		11–16, 28–33, 45–47
96		12–16, 29–33, 46–48
97		13–16, 30–33, 47–48
98		14–16, 31–33, 48–49
99		15–16, 32–33, 49
100		16, 33, 50

$\mathrm{LS}[17](4, k, v)$

None of the 594 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

$\mathrm{LS}[17](5, k, v)$

None of the 504 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

$\mathrm{LS}[17](6, k, v)$

None of the 420 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

$\mathrm{LS}[19](2, k, v)$

488 of 834 admissible parameter sets exist, 346 are open.

v	exists, k =	open, k =
21	3–4	5–10
22	4, **5**	6–11
23	5–10	11
24	6–10	11–12

v	exists, k =	open, k =
25	7–11	12
26	8–11	12–13
27	9–11	12–13
28	10–11	12–14
29	11	12–14
30		12–15
31	13–15	
32	14–16	
33	15–16	
34	16–17	
35	17	
36	18	
40	3–4	5–18
41	4, 12–14, 18	5–11, 15–17
42	13–14	5–12, 15–18
43	9–14, 16–18	6–8, 15
44	10–14, 17–18	7–9, 15–16
45	11–14, 18	8–10, 15–17
46	12–14	9–11, 15–18
47	10–18	
48	11–18	
49	12–18	
50	13–18	
51	14–18	
52	15–18	
53	16–18	
54	17–18	
55	18	
59	3–18, 22–28	29
60	4–18, 23–28	29–30
61	5–18, 24–28	29–30
62	6–18, 25–28	29–31
63	7–18, 26–28	29–31
64	8–18, 27–29	30–32
65	9–18, 28–29	30–32
66	10–18, 29	30–33
67	11–18, 32	30–31, 33
68	12–18	31–34
69	13–18	32–34
70	14–18	33–35
71	15–18, 34	35
72	16–18	35–36
73	17–18	36
74	18	37
78	3–4, 17–18	5–16, 22–37
79	4–5, 8, 11, 17–18, 23, 29, 32, 35	6–7, 9–10, 12–16, 24–28, 30–31, 33–34, 36–37
80	5–10, 13–18	11–12, 24–37
81	6–10, 14–18, 34	11–13, 25–33, 35–37
82	7–11, 14–18	12–13, 26–37
83	8–18, 27–37	
84	9–18, 28–37	
85	10–18, 29–37	
86	11–18, 30–37	
87	12–18, 31–37	
88	13–18, 32–37	
89	14–18, 33–37	
90	15–18, 34–37	
91	16–18, 35–37	
92	17–18, 36–37	
93	18, 37	
97	3–4, 35, 47	5–18, 22–34, 36–37, 41–46, 48
98	4, 12–14, 18	5–11, 15–17, 23–37, 42–49
99	13–14	5–12, 15–18, 24–37, 43–49
100	9–14, 16–18	6–8, 15, 25–37, 44–50

$\mathrm{LS}[19](3, k, v)$

87 of 714 admissible parameter sets exist, 627 are open.

v	exists, k =	open, k =
22		4–11
23	**5**	6–11

v	exists, k =	open, k =
24	7, 10	6, 8–9, 11–12
25		7–12
26		8–13
27		9–13
28	11	10, 12–14
29		11–14
30		12–15
31		13–15
32	14–16	
33	15–16	
34	16–17	
35	17	
36	18	
41		4–18
42	13	5–12, 14–18
43		6–18
44	10, 12–13, 17–18	7–9, 11, 14–16
45	13, 18	8–12, 14–17
46		9–18
47		10–18
48	11–15, 17–18	16
49	12–15, 18	16–17
50	13–15	16–18
51	14–15	16–18
52	15	16–18
53		16–18
54	17	18
55		18
60	4, 7–9, 11–17, 23, 27	5–6, 10, 18, 24–26, 28–30
61	8–9, 12–17	5–7, 10–11, 18, 24–30
62	9, 13–17	6–8, 10–12, 18, 25–31
63	14–17	7–13, 18, 26–31
64	15–17	8–14, 18, 27–32
65	16–17	9–15, 18, 28–32
66	17	10–16, 18, 29–33
67		11–18, 30–33
68	15	12–14, 16–18, 31–34
69		13–18, 32–34
70		14–18, 33–35
71		15–18, 34–35
72		16–18, 35–36
73		17–18, 36
74		18, 37
79		4–18, 23–37
80		5–18, 24–37
81		6–18, 25–37
82		7–18, 26–37
83		8–18, 27–37
84	9, 11, 13, 15, 17, 29, 31, 33, 37	10, 12, 14, 16, 18, 28, 30, 32, 34–36
85		10–18, 29–37
86		11–18, 30–37
87		12–18, 31–37
88		13–18, 32–37
89		14–18, 33–37
90		15–18, 34–37
91		16–18, 35–37
92		17–18, 36–37
93		18, 37
98		4–18, 23–37, 42–49
99		5–18, 24–37, 43–49
100		6–18, 25–37, 44–50

$\mathrm{LS}[19](4, k, v)$

None of the 600 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

$\mathrm{LS}[19](5, k, v)$

None of the 495 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[19](6, k, v)

None of the 396 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[23](2, k, v)

244 of 878 admissible parameter sets exist, 634 are open.

v	exists, k =	open, k =
25	3–4, 11	5–10, 12
26	4	5–13
27	5, 7–8, 10–11	6, 9, 12–13
28	8, 11	6–7, 9–10, 12–14
29	10–11	7–9, 12–14
30	11	8–10, 12–15
31	9–15	
32	10–16	
33	11–16	
34	12–17	
35	13–17	
36	14–18	
37	15–18	
38	16–19	
39	17–19	
40	18–20	
41	19–20	
42	20–21	
43	21	
44	22	
48	3–4	5–22
49	4, 18–19	5–17, 20–22
50	19	5–18, 20–22
51		6–22
52		7–22
53	9–11, 15–22	8, 12–14
54	10–11, 16–22	9, 12–15
55	11, 17–22	10, 12–16
56	18–22	11–17
57	19–22	12–18
58	20–22	13–19
59	14–22	
60	15–22	
61	16–22	
62	17–22	
63	18–22	
64	19–22	
65	20–22	
66	21–22	
67	22	
71	3–4, 9, 12–13, 17–19, 27–29, 32–34	5–8, 10–11, 14–16, 20–22, 26, 30–31, 35
72	4, 13, 18–19, 28–29, 33–34	5–12, 14–17, 20–22, 27, 30–32, 35–36
73	11, 14, 19, 29, 34–35	5–10, 12–13, 15–18, 20–22, 28, 30–33, 36
74	35	6–22, 29–34, 36–37
75		7–22, 30–37
76		8–22, 31–38
77		9–22, 32–38
78		10–22, 33–39
79	11, 17, 20, 35, 38	12–16, 18–19, 21–22, 34, 36–37, 39
80		12–22, 35–40
81	19, 22, 38	13–18, 20–21, 36–37, 39–40
82		14–22, 37–41
83	15–22, 38–40	41
84	16–22, 39–40	41–42
85	17–22, 40	41–42
86	18–22	41–43
87	19–22	42–43
88	20–22	43–44
89	21–22	44
90	22	45
94	3–4	5–22, 26–45
95	4	5–22, 27–45
96		5–22, 28–45
97	35, 38	6–22, 29–34, 36–37, 39–45
98		7–22, 30–45

v	exists, $k =$	open, $k =$
99		8–22, 31–45
100		9–22, 32–45

$\mathrm{LS}[23](3, k, v)$

81 of 778 admissible parameter sets exist, 697 are open.

v	exists, $k =$	open, $k =$
26		4–13
27		5–13
28	11	6–10, 12–14
29		7–14
30	11	8–10, 12–15
31		9–15
32	10–16	
33	11–16	
34	12–17	
35	13–17	
36	14–18	
37	15–18	
38	16–19	
39	17–19	
40	18–20	
41	19–20	
42	20–21	
43	21	
44	22	
49		4–22
50	19	5–18, 20–22
51		6–22
52		7–22
53		8–22
54	10–11, 17, 19, 21	9, 12–16, 18, 20, 22
55	11	10, 12–22
56		11–22
57		12–22
58		13–22
59		14–22
60	15–17, 19–22	18
61	16–17, 20–22	18–19
62	17, 21–22	18–20
63	22	18–21
64		19–22
65		20–22
66		21–22
67		22
72	4, 13, 19, 29	5–12, 14–18, 20–22, 27–28, 30–36
73		5–22, 28–36
74		6–22, 29–37
75		7–22, 30–37
76		8–22, 31–38
77		9–22, 32–38
78		10–22, 33–39
79		11–22, 34–39
80		12–22, 35–40
81		13–22, 36–40
82		14–22, 37–41
83		15–22, 38–41
84	17, 19, 39	16, 18, 20–22, 40–42
85		17–22, 40–42
86		18–22, 41–43
87		19–22, 42–43
88		20–22, 43–44
89		21–22, 44
90		22, 45
95		4–22, 27–45
96		5–22, 28–45
97		6–22, 29–45
98		7–22, 30–45
99		8–22, 31–45
100		9–22, 32–45

LS[23](4, k, v)

None of the 682 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[23](5, k, v)

None of the 592 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[23](6, k, v)

None of the 506 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[29](2, k, v)

All 905 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

LS[29](3, k, v)

124 of 827 admissible parameter sets exist, 703 are open.

v	exists, k =	open, k =
32	5–6, 9–16	4, 7–8
33	6, 10–16	5, 7–9
34	11–17	6–10
35	12–17	7–11
36	13–18	8–12
37	14–18	9–13
38	12–13, 15–19	10–11, 14
39	13, 16–19	11–12, 14–15
40	17–20	12–16
41	18–20	13–17
42	19–21	14–18
43	20–21	15–19
44	17–22	16
45	18–22	17
46	19–23	18
47	20–23	19
48	20–24	
49	21–24	
50	22–25	
51	23–25	
52	24–26	
53	25–26	
54	26–27	
55	27	
56	28	
61		4–28
62		5–28
63		6–28
64		7–28
65		8–28
66		9–28
67		10–28
68	15, 19, 21, 25	11–14, 16–18, 20, 22–24, 26–28
69		12–28
70		13–28
71		14–28
72	19	15–18, 20–28
73		16–28
74		17–28
75		18–28
76		19–28
77		20–28
78		21–28
79		22–28
80		23–28
81		24–28
82		25–28
83		26–28
84	27	28
85		28
90	4, 45	5–28, 33–44

v	exists, $k =$	open, $k =$
91		5–28, 34–45
92	46	6–28, 35–45
93		7–28, 36–46
94	47	8–28, 37–46
95		9–28, 38–47
96	48	10–28, 39–47
97		11–28, 40–48
98	49	12–28, 41–48
99		13–28, 42–49
100	50	14–28, 43–49

$\mathrm{LS}[29](4, k, v)$

1 of 751 admissible parameter sets exist, 750 are open.

v	exists, $k =$	open, $k =$
33	10	5–9, 11–16
34		6–17
35		7–17
36		8–18
37		9–18
38		10–19
39		11–19
40		12–20
41		13–20
42		14–21
43		15–21
44		16–22
45		17–22
46		18–23
47		19–23
48		20–24
49		21–24
50		22–25
51		23–25
52		24–26
53		25–26
54		26–27
55		27
56		28
62		5–28
63		6–28
64		7–28
65		8–28
66		9–28
67		10–28
68		11–28
69		12–28
70		13–28
71		14–28
72		15–28
73		16–28
74		17–28
75		18–28
76		19–28
77		20–28
78		21–28
79		22–28
80		23–28
81		24–28
82		25–28
83		26–28
84		27–28
85		28
91		5–28, 34–45
92		6–28, 35–46
93		7–28, 36–46
94		8–28, 37–47
95		9–28, 38–47
96		10–28, 39–48
97		11–28, 40–48
98		12–28, 41–49

v	exists, k =	open, k =
99		13–28, 42–49
100		14–28, 43–50

LS[29](5, k, v)

None of the 679 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[29](6, k, v)

None of the 609 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[31](2, k, v)

268 of 847 admissible parameter sets exist, 579 are open.

v	exists, k =	open, k =
33	3	4–16
34		4–17
35		5–17
36		6–18
37	11–18	7–10
38	12–19	8–11
39	13–19	9–12
40	14–20	10–13
41	12–20	11
42	13–21	12
43	14–21	13
44	15–22	14
45	16–22	15
46	17–23	16
47	17–23	
48	18–24	
49	19–24	
50	20–25	
51	21–25	
52	22–26	
53	23–26	
54	24–27	
55	25–27	
56	26–28	
57	27–28	
58	28–29	
59	29	
60	30	
64	3, 5, 11, 13, 17, 23, 25, 29	4, 6–10, 12, 14–16, 18–22, 24, 26–28, 30
65		4–30
66		5–30
67	6–8, 14–21, 24–26, 29	9–13, 22–23, 27–28, 30
68	7–8, 15–21, 25–26	9–14, 22–24, 27–30
69	8, 16–21, 26	9–15, 22–25, 27–30
70	17–21	9–16, 22–30
71	12–13, 17–21, 23–24, 27–29	10–11, 14–16, 22, 25–26, 30
72	13, 18–21, 24, 28–29	11–12, 14–17, 22–23, 25–27, 30
73	14, 19–21, 23, 26, 29	12–13, 15–18, 22, 24–25, 27–28, 30
74	20–21	13–19, 22–30
75	21	14–20, 22–30
76		15–30
77		16–30
78		17–30
79	20, 23, 29	18–19, 21–22, 24–28, 30
80		19–30
81	22–23	20–21, 24–30
82	23	21–22, 24–30
83	22–30	
84	23–30	
85	24–30	
86	25–30	
87	26–30	
88	27–30	
89	28–30	

v	exists, $k =$	open, $k =$
90	29–30	
91	30	
95	3–4	5–30, 34–47
96	4	5–30, 35–48
97	38, 47	5–30, 36–37, 39–46, 48
98		6–30, 37–49
99		7–30, 38–49
100		8–30, 39–50

$\mathrm{LS}[31](3, k, v)$

107 of 763 admissible parameter sets exist, 656 are open.

v	exists, $k =$	open, $k =$
34		4–17
35		5–17
36		6–18
37		7–18
38	12–13, 15–19	8–11, 14
39	13, 16–19	9–12, 14–15
40	17–20	10–16
41	18–20	11–17
42	13, 19–21	12, 14–18
43	20–21	13–19
44	17–22	14–16
45	18–22	15–17
46	19–23	16–18
47	20–23	17–19
48	18–24	
49	19–24	
50	20–25	
51	21–25	
52	22–26	
53	23–26	
54	24–27	
55	25–27	
56	26–28	
57	27–28	
58	28–29	
59	29	
60	30	
65		4–30
66		5–30
67		6–30
68	7, 15, 19, 21, 25	8–14, 16–18, 20, 22–24, 26–30
69		8–30
70		9–30
71		10–30
72	13, 19, 29	11–12, 14–18, 20–28, 30
73		12–30
74		13–30
75		14–30
76		15–30
77		16–30
78		17–30
79		18–30
80		19–30
81		20–30
82		21–30
83		22–30
84	23, 25, 27, 29	24, 26, 28, 30
85		24–30
86		25–30
87		26–30
88		27–30
89		28–30
90		29–30
91		30
96	4	5–30, 35–48
97		5–30, 36–48
98		6–30, 37–49
99		7–30, 38–49

v	exists, k =	open, k =
100		8–30, 39–50

LS[31](4, k, v)

None of the 681 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[31](5, k, v)

None of the 603 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[31](6, k, v)

None of the 527 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[37](2, k, v)

475 of 856 admissible parameter sets exist, 381 are open.

v	exists, k =	open, k =
39	3	4–19
40		4–20
41	14, 18–19	5–13, 15–17, 20
42	19	6–18, 20–21
43	11, 17–20	7–10, 12–16, 21
44	18–20	8–17, 21–22
45	19–20	9–18, 21–22
46	20	10–19, 21–23
47	11–22	23
48	12–22	23–24
49	13–23	24
50	14–23	24–25
51	15–23	24–25
52	16–23	24–26
53	17–23	24–26
54	18–23	24–27
55	19–23	24–27
56	20–23	24–28
57	21–23	24–28
58	22–23	24–29
59	23–28	29
60	24–28	29–30
61	25–28	29–30
62	26–28	29–31
63	27–28	29–31
64	28–29	30–32
65	29	30–32
66		30–33
67	32	31, 33
68		32–34
69		33–34
70		34–35
71		35
72		36
76	3	4–36
77		4–36
78		5–36
79	8, 11, 17, 20, 23, 29, 32, 35	6–7, 9–10, 12–16, 18–19, 21–22, 24–28, 30–31, 33–34, 36
80		7–36
81	19, 22–23, 34	8–18, 20–21, 24–33, 35–36
82	23	9–22, 24–36
83	10–36	
84	11–36	
85	12–36	
86	13–36	
87	14–36	
88	15–36	
89	16–36	
90	17–36	
91	18–36	

v	exists, k =	open, k =
92	19–36	
93	20–36	
94	21–36	
95	22–36	
96	23–36	
97	24–36	
98	25–36	
99	26–36	
100	27–36	

$\mathrm{LS}[37](3, k, v)$

16 of 805 admissible parameter sets exist, 789 are open.

v	exists, k =	open, k =
40		4–20
41		5–20
42		6–21
43		7–21
44	19	8–18, 20–22
45		9–22
46		10–23
47		11–23
48	13, 15, 17, 19	12, 14, 16, 18, 20–24
49		13–24
50		14–25
51		15–25
52		16–26
53		17–26
54		18–27
55		19–27
56		20–28
57		21–28
58		22–29
59		23–29
60		24–30
61		25–30
62		26–31
63		27–31
64		28–32
65		29–32
66		30–33
67		31–33
68		32–34
69		33–34
70		34–35
71		35
72		36
77		4–36
78		5–36
79		6–36
80		7–36
81		8–36
82		9–36
83		10–36
84	11, 13, 15, 17, 19, 23, 25, 27, 29, 31, 33	12, 14, 16, 18, 20–22, 24, 26, 28, 30, 32, 34–36
85		12–36
86		13–36
87		14–36
88		15–36
89		16–36
90		17–36
91		18–36
92		19–36
93		20–36
94		21–36
95		22–36
96		23–36
97		24–36
98		25–36
99		26–36
100		27–36

LS[37](4, k, v)

None of the 755 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[37](5, k, v)

None of the 707 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[37](6, k, v)

None of the 660 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[41](2, k, v)

257 of 890 admissible parameter sets exist, 633 are open.

v	exists, k =	open, k =
43	3, 5, 11–13, 16–20	4, 6–10, 14–15, 21
44	12–13, 17–20	4–11, 14–16, 21–22
45	13, 18–20	5–12, 14–17, 21–22
46	19–20	6–18, 21–23
47	7–22	23
48	8–22	23–24
49	9–23	24
50	10–23	24–25
51	11–23	24–25
52	12–23	24–26
53	13–25	26
54	14–25	26–27
55	15–25	26–27
56	16–25	26–28
57	17–25	26–28
58	18–25	26–29
59	19–28	29
60	20–28	29–30
61	21–28	29–30
62	22–28	29–31
63	23–28	29–31
64	24–29	30–32
65	25–29	30–32
66	26–29	30–33
67	27–29, 32	30–31, 33
68	28–29	30–34
69	29	30–34
70		30–35
71	32–34	31, 35
72	33–34	32, 35–36
73	34–35	33, 36
74	35	34, 36–37
75		35–37
76		36–38
77		37–38
78		38–39
79		39
80		40
84	3	4–40
85		4–40
86		5–40
87		6–40
88		7–40
89	10, 14–15, 26–27, 30–31, 35, 38–39	8–9, 11–13, 16–25, 28–29, 32–34, 36–37, 40
90	15, 27, 31, 39	9–14, 16–26, 28–30, 32–38, 40
91		10–40
92		11–40
93		12–40
94		13–40
95		14–40
96		15–40
97	35, 38	16–34, 36–37, 39–40
98		17–40
99		18–40
100		19–40

LS[41](3, k, v)

74 of 833 admissible parameter sets exist, 759 are open.

v	exists, k =	open, k =
44	13, 17–20	4–12, 14–16, 21–22
45	18–20	5–17, 21–22
46	19–20	6–18, 21–23
47	20	7–19, 21–23
48	10–11, 13–15, 17–22	8–9, 12, 16, 23–24
49	11, 14–15, 18–22	9–10, 12–13, 16–17, 23–24
50	15, 19–23	10–14, 16–18, 24–25
51	20–23	11–19, 24–25
52	21–23	12–20, 24–26
53	22–23	13–21, 24–26
54	17, 19, 21, 23–25	14–16, 18, 20, 22, 26–27
55	24–25	15–23, 26–27
56	25	16–24, 26–28
57		17–28
58		18–29
59		19–29
60	20–23, 25–28	24, 29–30
61	21–23, 26–28	24–25, 29–30
62	22–23, 27–28	24–26, 29–31
63	23, 28	24–27, 29–31
64		24–32
65		25–32
66		26–33
67		27–33
68		28–34
69		29–34
70		30–35
71		31–35
72		32–36
73		33–36
74		34–37
75		35–37
76		36–38
77		37–38
78		38–39
79		39
80		40
85		4–40
86		5–40
87		6–40
88		7–40
89		8–40
90		9–40
91		10–40
92		11–40
93		12–40
94		13–40
95		14–40
96		15–40
97		16–40
98		17–40
99		18–40
100		19–40

LS[41](4, k, v)

None of the 777 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[41](5, k, v)

None of the 723 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[41](6, k, v)

None of the 670 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

LS[43](2, k, v)

287 of 862 admissible parameter sets exist, 575 are open.

v	exists, k =	open, k =
45	3	4–22
46		4–23
47	5–22	23
48	6–22	23–24
49	7–23	24
50	8–23	24–25
51	9–23	24–25
52	10–23	24–26
53	11–25	26
54	12–25	26–27
55	13–25	26–27
56	14–25	26–28
57	15–25	26–28
58	16–25	26–29
59	17–28	29
60	18–28	29–30
61	19–28	29–30
62	20–28	29–31
63	21–28	29–31
64	22–29	30–32
65	23–29	30–32
66	24–29	30–33
67	25–29, 32	30–31, 33
68	26–29	30–34
69	27–29	30–34
70	28–29	30–35
71	29, 32–34	30–31, 35
72	33–34	30–32, 35–36
73	34–35	31–33, 36
74	35	32–34, 36–37
75		33–37
76		34–38
77		35–38
78		36–39
79	38	37, 39
80		38–40
81		39–40
82		40–41
83		41
84		42
88	3	4–42
89	10, 14–15, 26–27, 30–31, 35, 38–39, 42	4–9, 11–13, 16–25, 28–29, 32–34, 36–37, 40–41
90	15, 27, 31, 39	5–14, 16–26, 28–30, 32–38, 40–42
91		6–42
92		7–42
93		8–42
94		9–42
95		10–42
96		11–42
97	35, 38	12–34, 36–37, 39–42
98		13–42
99		14–42
100		15–42

LS[43](3, k, v)

60 of 802 admissible parameter sets exist, 742 are open.

v	exists, k =	open, k =
46		4–23
47		5–23
48	6–7, 10–11, 13–15, 17–22	8–9, 12, 16, 23–24
49	7, 11, 14–15, 18–22	8–10, 12–13, 16–17, 23–24
50	15, 19–23	8–14, 16–18, 24–25
51	20–23	9–19, 24–25
52	21–23	10–20, 24–26
53	22–23	11–21, 24–26
54	17, 19, 21, 23, 25	12–16, 18, 20, 22, 24, 26–27

v	exists, k =	open, k =
55		13–27
56		14–28
57		15–28
58		16–29
59		17–29
60	19–23, 26–27	18, 24–25, 28–30
61	20–23, 27	19, 24–26, 28–30
62	21–23	20, 24–31
63	22–23	21, 24–31
64	23	22, 24–32
65		23–32
66		24–33
67		25–33
68		26–34
69		27–34
70		28–35
71		29–35
72		30–36
73		31–36
74		32–37
75		33–37
76		34–38
77		35–38
78		36–39
79		37–39
80		38–40
81		39–40
82		40–41
83		41
84		42
89		4–42
90		5–42
91		6–42
92		7–42
93		8–42
94		9–42
95		10–42
96		11–42
97		12–42
98		13–42
99		14–42
100		15–42

$\mathrm{LS}[43](4, k, v)$

None of the 743 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

$\mathrm{LS}[43](5, k, v)$

None of the 686 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

$\mathrm{LS}[43](6, k, v)$

None of the 630 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.